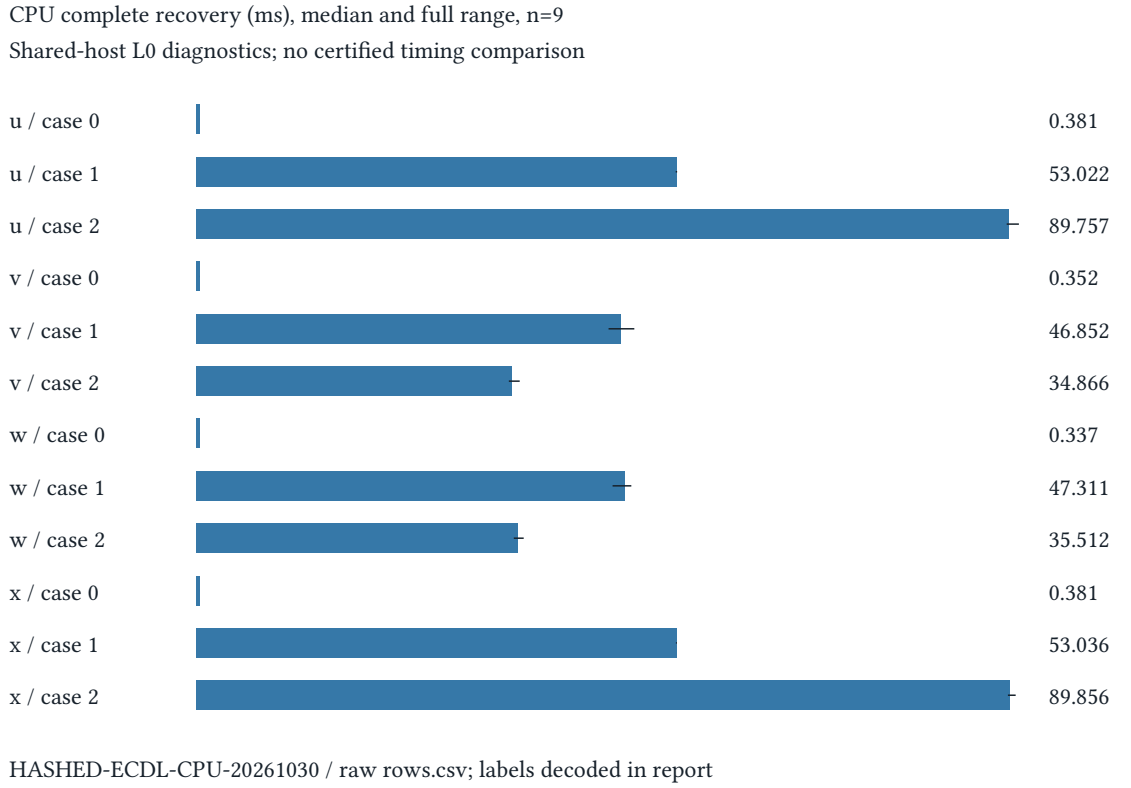
CPU complete recovery: initial cache

Observations only; no hypothesis promotion or certified latency claim. Run: `hashed-ecdl-cpu-panel1-v3-20261030`, October 29, 2026 PDT.

The original CPU oracle (u), reusable tables with batch 64 (v), batch 1 ablation (w), and identical reference repeat (x) ran on one software thread in an EKS job reserving two vCPUs and 4 GiB on AMD EPYC 9R45. The main-built runtime image is distinct from the pinned standalone CPU instrument. All clocks are L0.

Fixture	Original u, ms	Cached v, ms	A/A x, ms
8-bit, 3 slots	0.380702	0.352192	0.380672
16-bit, 3 slots	53.021651	46.852160	53.035951
16-bit, 4 slots	89.756843	34.865870	89.855764

Medians of nine measured repetitions; one warmup per arm and fixture. Exact curves, public known-answer targets and seeds 111/222/333 are in the protocol.



All 120 requested executions completed (108 measured, 12 warmups), with byte-identical complete certificates across arms. Scalar recovery, checked group replay and deliberately corrupted scalar rejection passed. The interval includes curve and target construction, all setup and trials, relation collection, elimination, internal replay, second replay and serialization. Process launch, archive I/O and the negative control are excluded. Original zero table/inversion counters are uninstrumented/inapplicable, not zero arithmetic.

Two staging jobs failed before any measurement: bucket HeadObject 403, then missing tar. Both receipts are retained; protocol versions 1–3 document the transport corrections. Final artifacts used scoped presigned URLs and standard archive handling, with the native benchmark and analyzer unchanged.

Source `b02a2f2dd130`; full source/binary hashes: `results/build.json`. Protocol hash: `83ad81c97c9919b34f918bdfa0d3d1a6d645512f4a6dfa4f97f16a16804bfcea`. Raw rows, full certificates, stage ranges and checked export: `results/`. Matched rho and fastest external comparisons are absent; their gaps are unknown. This generic decomposition result is not Boolean tensor contraction or evidence of asymptotic improvement. A later panel separately tests a parameter-only key.